Event-Level Financial Sentiment Analysis for English Financial Texts

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Abstract

Event-level financial sentiment analysis (EFSA) isolates sentiment toward specific corporate events to provide granular signals for event-driven trading and risk assessment. We adapt Chen et al. (2024)'s EFSA framework to English financial texts by enriching FiQA-2018 data (1,174 instances) with company, industry, and hierarchical event labels, implementing a five-hop chain-of-thought approach using Mistral-7B-Instruct across zero-shot, few-shot, LoRA-tuned, and LoRA-tuned with zero-shot configurations. Our LoRA-tuned model achieves 0.97 weighted-F1 for individual sentiment classification, substantially outperforming fine-tuned DistilRoBERTa's 0.77, but suffers from severe error propagation in multi-step reasoning (0.39 F1 on complete EFSA). Results demonstrate the promise of parameter-efficient fine-tuning for financial domain adaptation and the fundamental challenges of sequential reasoning in domain-specific NLP tasks.

1 Introduction

Recent advances in large language models have established financial sentiment analysis (FSA) as an increasingly important research domain at the intersection of natural language processing (NLP) and finance. Financial analysts routinely examine textual data alongside quantitative metrics across diverse applications such as financial forecasting, credit scoring, company valuation, and volatility modeling. This textual data encompasses various formats, from long-form corporate communications and published news articles to short-form blog and social media posts.

However, given the volatility of financial markets, simply identifying overall sentiment associated with a piece of text is insufficient. Understanding the underlying events that drive market fluctuations is crucial, as these events provide specific and actionable insights for financial applications. Despite this need, limited research has attempted to extract granular event-level information that aligns with financial analyst workflows.

Traditional FSA approaches suffer from a fundamental limitation: they conflate multiple events within single documents, obscuring the distinct impact of discrete corporate actions. Event-level analysis addresses this gap by offering three key advantages. First, it provides improved granularity by isolating sentiment for specific events rather than aggregating across entire documents. Second, it delivers increased practical value for event-driven trading algorithms that require precise sentiment signals tied to particular corporate actions. Third, it aligns with established financial analyst workflows, which focus on discrete market updates rather than overall document tone.

This project develops an English event-level financial sentiment analysis system based on the Event-Level Financial Sentiment Analysis (EFSA) framework of Chen et al. (2024). Rather than analyzing sentiment at the document or sentence level, our system extracts sentiment at the granular event level. The system identifies specific corporate events at both coarse-grained and fine-grained levels from financial news headlines and tweets, then classifies sentiment regarding their impact on individual stock values.

2 Background

Financial sentiment analysis has evolved significantly since Loughran and McDonald (2011) first pioneered domain-specific FSA by developing specialized financial word lists to analyze 10-K filings. This seminal work established the necessity for domain-specific approaches towards sentiment analysis. Based on this foundation, researchers have systematically expanded FSA in multiple analytical granularities, from sentence-level (Malo et al., 2014), aspect-level (Yang and Rosenfeld,

2018), to document-level (Araci, 2019) approaches to FSA.

The introduction of language models has fundamentally reshaped FSA capabilities, with transformer-based models achieving remarkable improvements in evaluation metrics by capturing nuanced domain-specific contexts and features (Dmonte et al., 2024). The introduction of domain-specific LLMs such as FinBERT has further advanced these capabilities by better capturing nuanced contexts within financial language and outperforming general-purpose models such as GPT-3.5 in downstream predictive tasks (Balaneji, 2024).

Despite these LLM advances, traditional FSA approaches continue to suffer from a fundamental limitation. Most existing studies focus on predicting sentiments while neglecting the analysis of distinct events within financial texts. Recent work on aspect-based financial sentiment analysis attempts to address granularity issues by demonstrating that fine-grained aspect-based sentiment analysis can assign sentiment polarity scores to specific entities within financial texts (Consoli et al., 2022). The EFSA framework introduced by Chen et al. (2024) addresses this gap by extracting quintuples consisting of (company, industry, coarse-grained event, fine-grained event, sentiment) from Chinese financial texts, recognizing that financial texts often contain multiple events with potentially different sentiment implications.

Current FSA research continues to grapple with challenges such as data quality (Sinha et al., 2022), model interpretability (Xing et al., 2020), and integration of sentiment analysis with quantitative financial models. The field is increasingly moving toward more sophisticated multimodal approaches, which combine textual sentiment analysis with quantitative technical indicators and market data (Chen et al., 2023; Zhang et al., 2023), suggesting that future FSA frameworks will incorporate even more advanced architectures to handle the complexities of financial texts and temporal sensitivity.

This research builds upon established foundation of FSA while addressing the specific gap in event-level granularity for English corpora. By extending the EFSA framework, our work contributes to providing granular and actionable insights that financial analysts and individual traders require for event-driven trading strategies and risk assessment procedures.

3 Methods

3.1 Problem Formulation and Evaluation

We formulate EFSA as a sequential quintuple extraction task. Given a financial text instance x, our model must predict $(c, i, e_{coarse}, e_{fine}, s)$, where c is company name, i is GICS industry sector $\in$ {74 sectors}, $e_{coarse} \in$ {7 superclasses} is coarse-grained event, $e_{fine} \in$ {34 subclasses} is fine-grained event, and $s \in$ {NEG, NEU, POS} is sentiment polarity.

Unlike traditional sentiment analysis, our approach recognizes that financial sentiment requires contextual understanding. For example, "AstraZeneca Explores Potential Deal With Acerta for Cancer Drug" extracts (*AstraZeneca*, *Pharmaceuticals*, *Business Operations*, *Initiating Cooperation*, *POS*), where industry context is crucial—the same type of deal may carry different implications across different business sectors.

Our evaluation framework measures progressive tuple-level performance: sentiment-only accuracy (s), entity-level FSA (c, i, s), coarse-grained EFSA (c, i, e_{coarse}, s), fine-grained EFSA (c, i, e_{fine}, s), and complete EFSA $(c, i, e_{coarse}, e_{fine}, s)$. We target weighted-F1 > 0.80 for individual classification components and weighted-F1 > 0.70 for tuple-level predictions.

3.2 Dataset Construction and Processing

Given the novelty of EFSA, available annotated data for the English corpus is limited. We adapt the FiQA Challenge dataset from 2018, specifically utilizing Task 1 data from the FiQA Sentiment Classification repository (TheFinAI, 2023), containing 1,173 text instances comprising 498 news headlines and 675 financial tweets. Each instance includes three key attributes: target (referenced company), aspect (hierarchical categorization system), and sentiment score (ranging from -1 to 1).

Sentiment Label Mapping. We transform continuous sentiment scores into discrete categorical labels where instances with score > 0 receive POS labels, score < 0 receive NEG labels, and scores = 0 receive NEU labels.

Financial Entity Normalization. We address unique challenges in financial entity recognition, including inconsistent company name abbreviations, ticker symbol variations, and informal social media references across heterogeneous text sources.

Hierarchical Aspect Mapping. We enhance the original two-level hierarchical structure by map-

ping these aspects to the detailed event taxonomies established by Chen et al. (2024). This transformation creates more precise event categories that effectively capture the semantics of the financial domain. For instance, Stock/Price Action becomes coarse-event Stock Affairs and fine-event Stock Price Movement.

Industry Integration. We enrich the dataset through integration of the yfinance API for GICS-aligned industry classifications.

The enhanced dataset contains seven key columns: Sentence, Type, Company, Industry, Coarse-Grained Event, Fine-Grained Event, and Sentiment. The dataset is partitioned into training (822 instances), testing (234 instances), and validation (117 instances) sets.

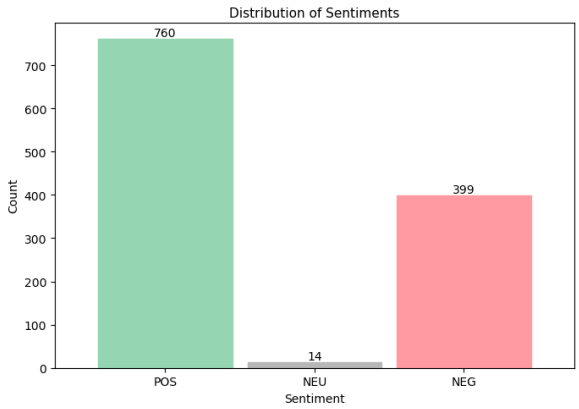

Figure 1: Label distribution for the Sentiment column in the adapted FiQA dataset.

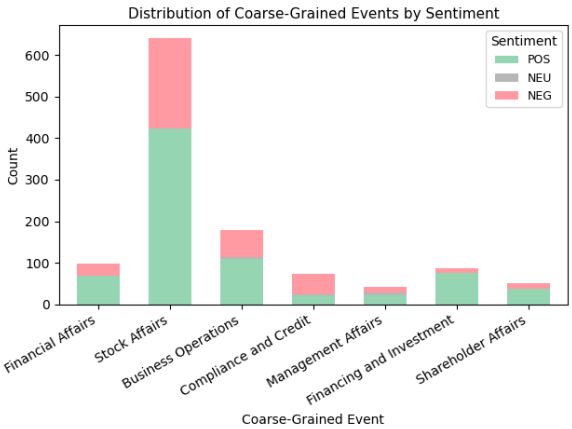

Figure 2: Data distribution by Coarse-Grained Event categories and associated Sentiment labels.

3.3 Dataset Distribution

The resulting dataset demonstrates a pronounced sentiment class imbalance where positive labels represent approximately 65% of instances, negative labels constitute 34%, and neutral labels account for only 1%. This distribution reflects financial communication dynamics where stakeholders rarely maintain neutral positions in market discourse. Therefore, we preserve this authentic distribution instead of applying sampling techniques that could introduce artificial biases.

Coarse-grained event distribution reveals that Stock Affairs dominates the taxonomy, representing approximately 50% of all instances, while other categories, such as Financial Affairs, Corporate Affairs, and Market Affairs, each account for 5-15% of samples. This concentration reflects the market-centric nature of financial discourse across news and social media platforms.

3.4 Baseline Model

To establish a performance floor for event-level financial sentiment analysis, we evaluate a diverse set of baseline models, spanning classical sequence architectures and modern pretrained transformers. Classical architectures—convolutional neural networks (CNN), recurrent neural networks (RNN), and long short-term memory networks (LSTM)—serve as strong references for learned feature extraction and sequential modeling. In parallel, we include two transformer-based variants: DistilBERT-SST-2 (a distilled BERT fine-tuned on movie-review data) and BERT-base-uncased, to quantify the benefits of large-scale pretraining. Finally, we assess DistilRoBERTa in a zero-shot setting: a distilled version of RoBERTa pretrained on financial-news sentiment data but not further tuned on our task.

3.5 Fine-Tuning DistilRoBERTa

To adapt DistilRoBERTa more closely to our event-level sentiment classification task, we performed task-specific fine-tuning using our training set (822 samples) and evaluated on the held-out validation set (234 samples). Key steps and configuration details are as follows:

Label Mapping. Labels {NEG, NEU, POS} were mapped to integer IDs in alphabetical order.

Tokenization. Each sentence was tokenized with the pretrained DistilRoBERTa tokenizer, using a maximum sequence length of 128 and standard padding/truncation.

Training Configuration. We fine-tuned using the AdamW optimizer with an initial learning rate of 5×10^{-5}, a per-device batch size of 16, and

weight decay of 0.01. A linear learning-rate decay schedule was applied with warmup steps set to 10% of the total training steps. Training proceeded for up to three epochs, with early stopping on the weighted-average F1 score (patience of one epoch) to select the best checkpoint.

Optimization and Checkpoint. We employed AdamW with linear learning-rate decay. The checkpoint achieving the highest macro-averaged F1 on the validation set was retained as the final model.

Evaluation. After fine-tuning, we generated predictions on the validation split and computed precision, recall, and F1-score for each class using scikit-learn.

3.6 Five-Hop Chain-of-Thought Model

Our final model framework implements a five-hop Chain-of-Thought approach using the Mistral-7B-Instruct autoregressive LLM. This methodology extends Chen et al. (2024)'s four-hop framework by eliminating external dependency on industry lookup tables, performing dynamic industry classification as an integral component of the end-to-end reasoning pipeline.

The five-hop approach follows a progressive reasoning structure that mirrors financial analysts' workflows:

Company Extraction. The system extracts all mentioned company names, addressing financial entity recognition challenges including informal social references and ticker symbol variations.

Industry Classification. Using textual context and extracted company names, the LLM classifies each company into one of 74 GICS industry sectors.

Coarse-Event Classification. The system processes the financial text along with the company name and industry classification to determine a corresponding coarse-grained event from 7 defined superclasses.

Fine-Event Classification. With coarse-event classification constraining available options, the LLM selects the corresponding fine-grained event from a list of relevant subclasses.

Sentiment Classification. Drawing on complete contextual information through the quadruple (company, industry, coarse-grained event, fine-grained event), the LLM classifies sentiment polarity.

Each reasoning step employs optimized temperature settings (0.05-0.2) and tuned prompts with strict output constraints requiring responses from predefined lists. The sequential design leverages intermediate predictions for improved accuracy while introducing error propagation risks, with industry classification serving as a critical contextual bridge for event interpretation and sentiment assessment.

We evaluate four configurations: **Zero-Shot** uses base prompts without examples, **Few-Shot** incorporates 3 examples per task, **LoRA-Tuned** fine-tunes on training and validation data, and **LoRA-Tuned with Zero-Shot** applies the fine-tuned model using zero-shot prompts to assess transfer learning capabilities.

LoRA fine-tuning uses Mistral-7B-Instruct-v0.3 with 4-bit quantization, r=8, alpha=16, dropout=0.1, targeting `q_proj` and `v_proj` layers. Training employs gradient accumulation, learning rate=1×10^{-5}, and early stopping based on weighted-F1, enabling parameter-efficient domain adaptation.

4 Results and Discussion

4.1 Baseline Model

Classical Sequence Models. CNN and LSTM achieve the highest classical accuracy (0.77) and weighted-F1 (0.49–0.51), demonstrating that both local context aggregation and gated recurrence capture key sentiment cues in financial texts. The RNN baseline trails at 0.65 accuracy and 0.26 weighted-F1, reflecting the challenge of long-range dependency modeling without gating mechanisms. All three incur substantial inference latency (70–73 ms per sentence), which may hinder real-time applications.

Pretrained Transformers. DistilBERT-SST-2 processes each sentence in just 7.1 ms, achieving 0.62 accuracy and 0.58 weighted-F1—an order of magnitude faster than CNN/LSTM while delivering competitive performance. In contrast, BERT-base-uncased yields only 0.49 accuracy and 0.44 weighted-F1 at 11.3 ms, underscoring that generic movie-review pretraining does not directly transfer to financial domain semantics.

DistilRoBERTa. We adopt DistilRoBERTa, a smaller, faster variant of RoBERTa obtained by knowledge distillation and further pretrained on financial news, as our primary transformer baseline. Without any task-specific tuning, it achieves 0.50 accuracy and 0.44 macro-F1 with 60 ms latency, revealing a clear domain shift between financial news headlines and our event-level corpus.

These results, summarized in Table 1, highlight

Model	Sentiment Accuracy	Weighted-F1	Inference (ms/sent)
CNN	0.77	0.49	70.3
RNN	0.65	0.26	73.0
LSTM	0.77	0.51	70.8
DistilBERT-SST-2	0.62	0.58	7.1
BERT-base-uncased	0.49	0.44	11.3
DistilRoBERTa	0.50	0.44	60.0

Table 1: Performance comparison of various models on the event-level financial sentiment classification task

that, while classical models can reach moderate accuracy, their inference cost is high. Transformer variants, particularly those distilled and domain-adapted, offer superior throughput and strong accuracy, motivating our choice of DistilRoBERTa as the foundation for subsequent fine-tuning.

4.2 Fine-Tuned Baseline Model

Table 2 summarizes the classification report on the validation set after fine-tuning. Fine-tuning yields an overall accuracy of 0.84, with substantial gains in positive and negative classes. The neutral class remains challenging due to its low support (3 examples), indicating that future work should consider additional neutral-class augmentation or cost-sensitive learning to bolster performance on this minority class.

4.3 Five-Hop Chain-of-Thought

Our experiments with different CoT configurations-Zero-Shot, Few-Shot, LoRA-Tuned, and LoRA-Tuned with Zero-Shot prompting-reveal several key findings across individual classification tasks (Table 3). The LoRA-Tuned model consistently achieves the strongest performance, with particularly notable results in sentiment classification (0.97 weighted-F1) representing a substantial improvement over both baseline models and the fine-tuned DistilRoBERTa (0.77). The model also demonstrates robust performance on company identification (0.95 weighted-F1), industry classification (0.85 weighted-F1), and coarse event analysis (0.82 weighted-F1).

Zero-Shot vs. Few-Shot Performance. Surprisingly, Zero-Shot prompting outperforms Few-Shot across most tasks. This counterintuitive result may reflect the strong instruction-following capabilities of Mistral-7B-Instruct, where adding more examples may add more noise than helpful guidance.

Task Complexity and Performance Degradation. A clear pattern emerges where performance degrades as the number of classes within a category increases (Sentiment: 0.97 $\rightarrow$ Coarse Event: 0.82 $\rightarrow$ Fine Event: 0.72), suggesting that the LLM may struggle with fine-grained distinctions.

Domain Adaptation Effectiveness. The LoRA-tuned approach demonstrates promising results for adapting a general purpose autoregressive LLM to the financial domain. Despite a limited dataset, the model achieves substantial performance gains over baseline models, suggesting that parameter-efficient fine-tuning can effectively capture domain-specific contextual cues and terminology.

4.4 Limitations and Error Analysis

Our variant task performance falls substantially short of (Chen et al., 2024)'s original EFSA results, where their best model (Qwen-7B-Chat with 4-hop CoT) achieved 0.71 F1 on EFSA tasks compared to our 0.39 F1. This performance gap highlights several key methodological differences and limitations in our approach.

Error Propagation in Multi-Step Reasoning. Our 5-hop CoT framework exhibits severe error accumulation, with performance dropping from 0.97 F1 in individual sentiment tasks to 0.39 F1 in EFSA variants. This confirms findings from previous work that sequential reasoning chains are susceptible to cascading errors, with each additional step introducing approximately 15% performance degradation. The industry classification step, absent from the original EFSA framework, serves as an additional error source that compounds downstream prediction failures.

Dataset and Scale Limitations. Our small dataset (822 training samples) with severe class imbalances creates fundamental generalization challenges compared to (Chen et al., 2024)'s 50,000 samples dataset. With only 1% of samples in the NEU sentiment class and 35% in the NEG class, plus similar imbalances across event and entity categories, our model struggles to learn meaningful

Class	Precision	Recall	Weighted-F1	Support
Sentiment	0.78	0.76	0.77	80
Industry	0.00	0.00	0.00	3
Coarse-Grained Event	0.87	0.89	0.88	152
Accuracy	-	0.84	-	235
Macro Avg	0.55	0.55	0.55	235
Weighted Avg	0.83	0.84	0.83	235

Table 2: Classification report on the validation split after fine-tuning DistilRoBERTa.

	Weighted-F1			
Task	**Zero**	**Few**	**LoRA**	**LoRA-Zero**
Company	0.74	0.56	0.86	0.95
Industry	0.43	0.30	0.85	0.37
Coarse Event	0.83	0.65	0.82	0.73
Fine Event	0.48	0.44	0.72	0.44
Sentiment	0.81	0.82	**0.97**	0.83
EFSA	0.13	0.12	**0.39**	0.09
Coarse EFSA	0.35	0.22	0.51	0.32
Fine EFSA	0.13	0.12	0.46	0.09
Entity-Level FSA	0.37	0.24	**0.70**	0.36

Table 3: COT performance comparison of Zero-Shot, Few-Shot, LoRA and LoRA with Zero-Shot prompt methods.

semantic contexts to clearly differentiate between labels, particularly affecting the complex reasoning required for variant FSA tasks.

5 Conclusion

We have presented an end-to-end English EFSA framework that extends the original through a novel five-hop Chain-of-Thought approach, integrating company extraction, dynamic industry classification, hierarchical event mapping, and sentiment analysis within a unified reasoning pipeline. Our evaluation demonstrates the feasibility of applying EFSA to English financial texts, with LoRA-tuned Mistral-7B-Instruct achieving exceptional performance on individual classification tasks (0.97 weighted-F1 for sentiment, 0.95 for company identification), though the multi-step reasoning approach suffers from severe error propagation, with performance dropping to 0.39 weighted-F1 on complete EFSA tasks.

The research reveals critical insights including the counterintuitive superiority of zero-shot over few-shot prompting for instruction-following models and the effectiveness of LoRA tuning for financial domain adaptation despite limited training data. However, our analysis confirms that sequential reasoning chains are highly susceptible to cascading failures, with each additional reasoning step introducing approximately 15% performance degradation. Error analysis highlights fundamental challenges including severe class imbalances, dataset scale limitations, and the complexity of maintaining coherent reasoning across extended contexts.

Future work should address error propagation through joint optimization approaches, expand dataset sizes with stratified sampling, and explore end-to-end architectures that capture complex interdependencies without sequential reasoning chains. Moreover, our proposed five-hop CoT framework, despite current limitations, demonstrates that event-level financial sentiment analysis is indeed feasible for English texts and offers a promising foundation for synchronizing multi-faceted financial analysis within a single reasoning pipeline, paving the way for more sophisticated and context-aware event-level FSA in real-world financial applications.

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A Authors' Contributions

Coco Sun. (1) Performed data cleaning and feature extraction with FiQA dataset (2) Trained and tuned Mistral Chain-of-Thought Zero-Shot and Few-Shot models (3) Performed experimentation with other LLMs such as Llama and LoRA tuned models available on HuggingFace (4) Performed LoRA tuning using Mistral-7B-Instruct as base model (5) Performend model evaluation and comparison between Zero-Shot, Few-Shot, Lora-Tuned, and Lora with Zero-Shot models

Dezhi Yu. (1) Performed data preprocessing and sentiment analysis on EFSA financial news dataset with comprehensive label distribution analysis (POS, NEG, NEU) (2) Implemented and evaluated multiple transformer-based baseline models including BERT, DistilBERT, and DistilRoBERTa for financial sentiment classification (3) Conducted fine-tuning experiments using pre-trained DistilRoBERTa model (mrm8488/distilroberta-finetuned-financial-news-sentiment-analysis) as base model for domain-specific sentiment analysis (4) Performed comprehensive model evaluation and comparison using weighted F1-score, precision, recall metrics across different transformer architectures

Haoyue Zhang. (1) Established baseline performance using traditional deep learning models (CNN, RNN, LSTM) (2) Developed and evaluated a Chain-of-Thought framework that integrates Zero-shot, Few-shot, and LoRA fine-tuning approaches

B Coarse-Grained and Fine-Grained Event Taxonomy

Financial Affairs. Profit Announcement, Profit Forecast, Other Financial Affairs

Shareholder Affairs. Stock Holding Adjustment, Shareholder Pledge, Release of Pledge, Other Shareholder Affairs

Stock Affairs. Stock Price Movement, Equity Incentives & Employee Stock Ownership Plans, Stock Dividend, Stock Buyback, Stock Status, Restricted Shares Release, Other Stock Affairs

Business Operations. Product Dynamics, Capacity Changes, Initiating Cooperation, Technical

Quality Control, Qualification Changes, Government Subsidies, New Company Establishment, Institutional Research, Intellectual Property, Sales, Market Share Changes, Project Bidding, Project Dynamics, Other Business Operations Affairs

Compliance and Credit. Company Litigation, Rating Adjustment, Legal Affairs, Clarification Announcements, Regulatory Inquiries, Case Investigations, Administrative Penalties, Other Compliance and Credit Affairs.

Management Affairs. Employee Dynamics, Directors, Supervisors, and Senior Executives Dynamics

Financing and Investment. Company Listing, Mergers and Acquisitions, Investment Events, Stock Issuance, Financing and Margin Trading, Capital Flows, Other Financing and Investment Affairs